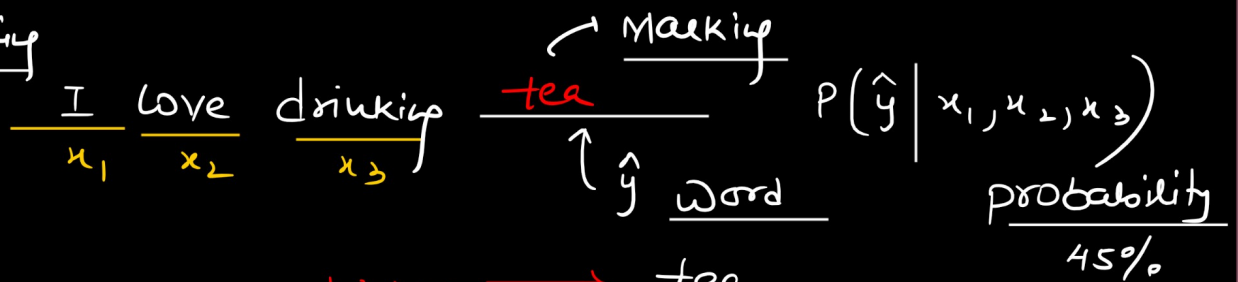


# Language Modeling

## Large Language Model

Predict the next token

### Pretraining



### Conditional

#### Probability

higher

probability

tea

Coffee

30%

$$P(\hat{y} | x_1, x_2, x_3) = \frac{P(A \cap B)}{P(B)}$$

A

B

Juice

15%

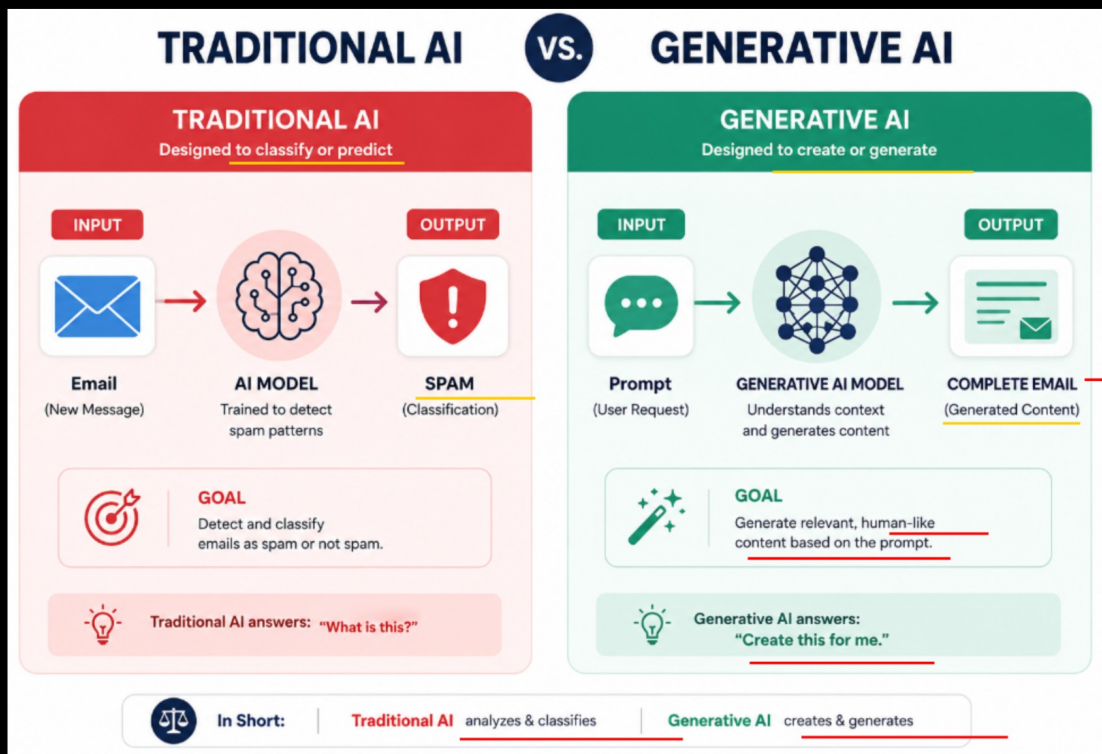
Water

10%

## How Generative AI Learns

From Massive Data to Unlimited Possibilities





generative  
the new  
content

## Discriminative Models

Classification

→ to distinguish b/w  
different classes

- Review → +ve/-ve
- Image → cat/dog
- transaction → fraud/genuine
- patient data → diabetic/normal

## Generative Models

→ to generate something  
new

↳ writing story

↳ generate image

↳ create music

## Discriminative vs Generative Models

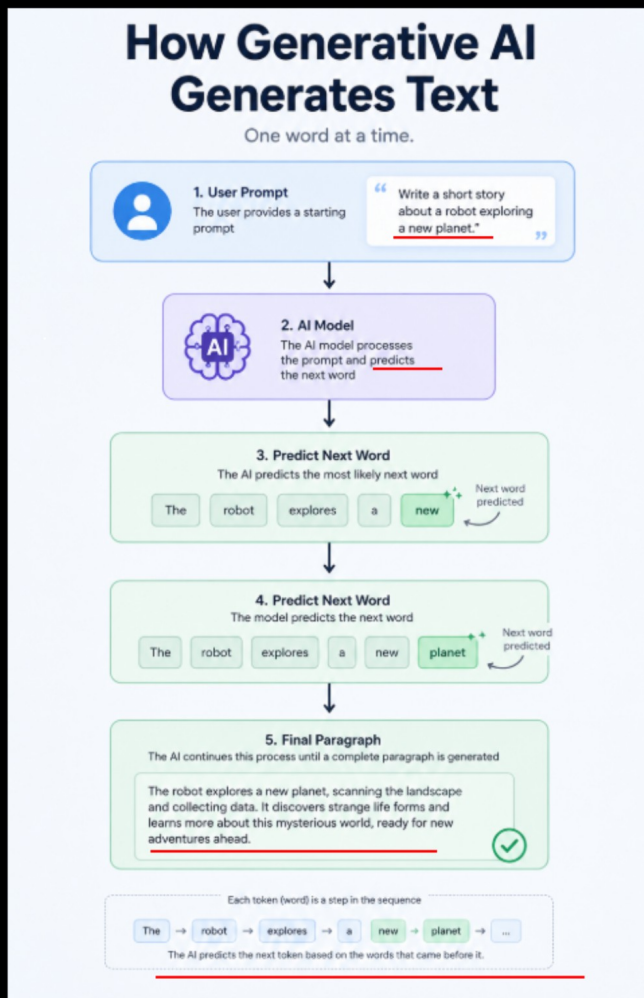
| Discriminative Models     | Generative Models                 |
|---------------------------|-----------------------------------|
| Predict labels            | Generate new data                 |
| Focus on classification   | Focus on content creation         |
| Learn decision boundaries | Learn data distribution           |
| Output is a prediction    | Output is newly generated content |

### Text Generation

User prompt → language model → Predict next  
Word (Repeat)

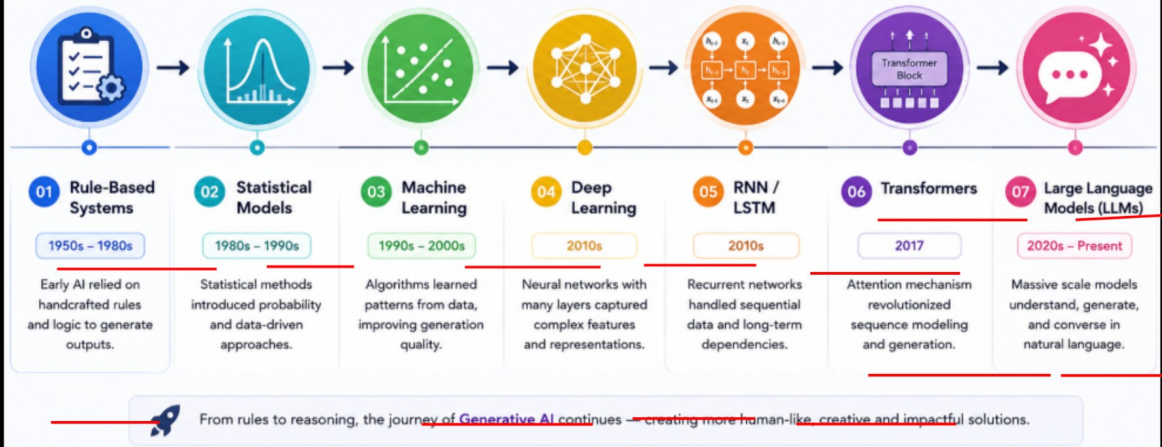
Complete

Response

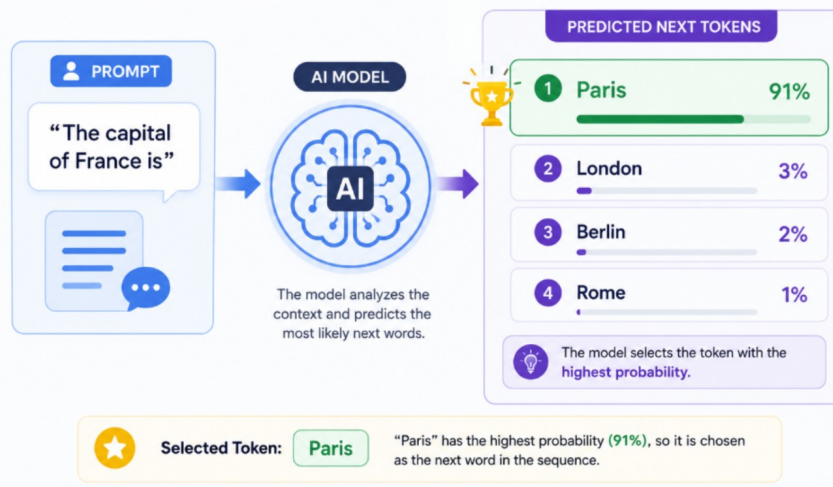


# Evolution of Generative AI

From Simple Rules to Intelligent Creation

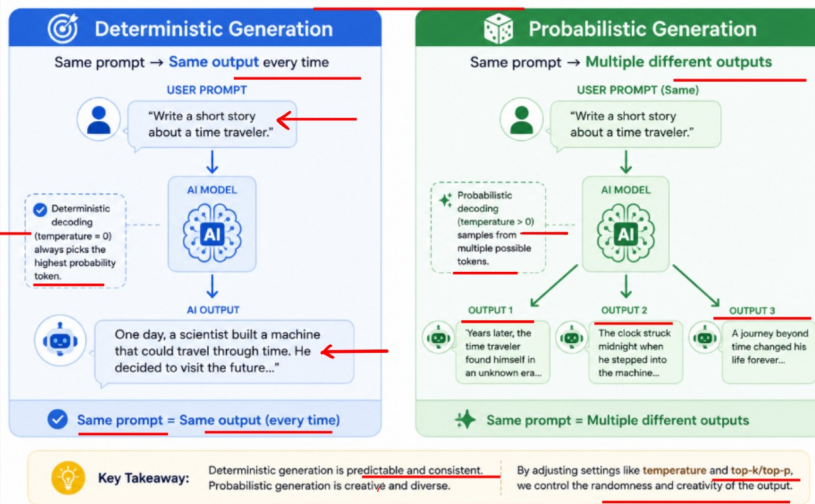


## How AI Predicts the Next Token



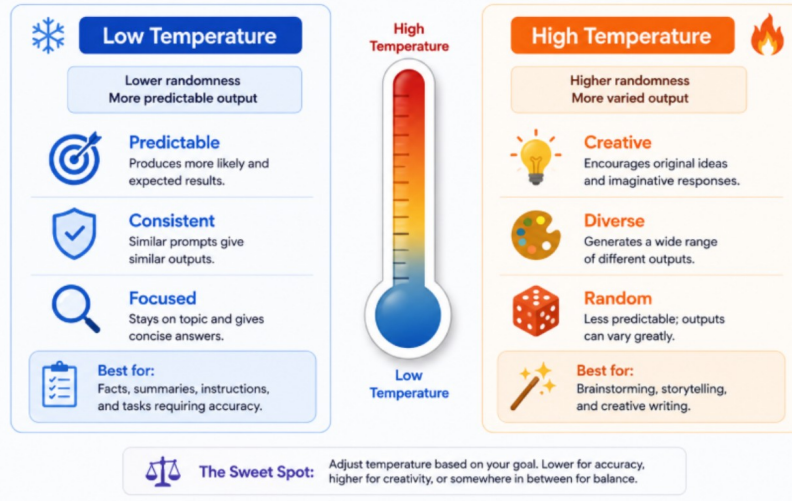
## Deterministic vs Probabilistic Generation

How AI can generate text in different ways



# Temperature in AI Text Generation

Temperature controls the randomness of the model's output.



→ Implementation → OpenAI API key

Gemini/Grok

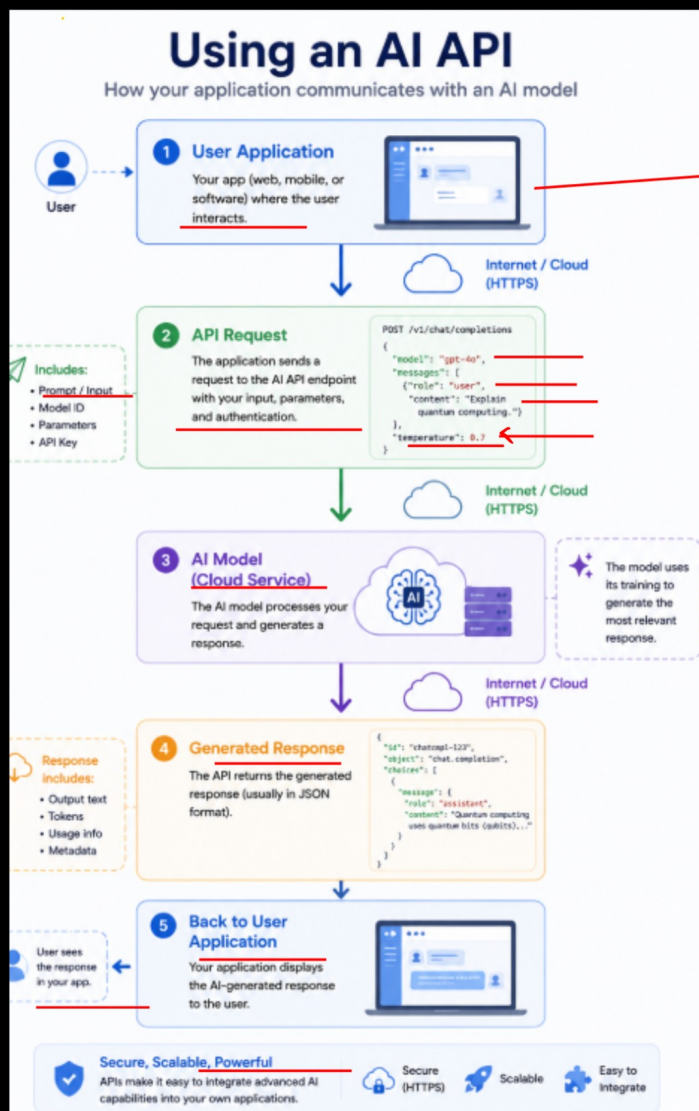
(free)

→ Post training →

SFT, RLHF

(Tuning)

(5\$)



Website, app

Post-training

RLHF → Reinforcement  
Learning With  
Human feedback

Reinforcement learning from human feedback (RLHF) is a training method that aligns large language models with human values, preferences, and intent.

